Trade the 8-K: Leadership Filings Offer No Tradable Option Edge; the Exit Surprise Is Priced Within a Day

Gator Quant Hacks 2026 · Massive bonus · Team: [team member names] · Code: [public GitHub URL] · `strategy_8k/`

1. Hypotheses (each committed to git before any option price for it was requested)

- **H-A, executive exits → cash-secured put** (`ceo/cfo/executive_officer_departure`, pooled; commit `dbbfe6c`). A vivid, ambiguous exit makes holders overpay for downside puts (demand pressure, Gârleanu et al. 2009), while mega-cap businesses rarely change overnight. So the put seller earns an event premium beyond the ordinary variance risk premium.
- **H-B, CFO appointment → cash-secured put** (workshop suggestion; `HYPOTHESIS_5.md`). Filling the CFO seat resolves financial-reporting uncertainty, but implied volatility mean-reverts with a lag, so sold puts collect the leftover premium. Pre-declared side tags: CEO appointment, and CFO departure (predicted *no* edge).

Spec = the notebook's defaults, so it matches what judges run: top-100 universe, 3–6m expiry bucket, entry at the close of the filing session, 5% OTM, fixed horizons, and the notebook's placebo (same tickers on ordinary days, 30 days or more from any event). Data: Massive REST API (hackathon key): option contracts as of the pre-event session (`/v3/reference/options/contracts`), spot from put-call parity, and daily contract bars (`/v2/aggs`, last trade, not NBBO mid). About 6–10k requests per window, cached locally and not shared (licensed). The starter's API cell and pipeline cells run **unchanged**; we only set the window and the event table (`exec_exits_8k.ipynb`, `EVENTS/START/END`).

2. Method: three windows and a star rule fixed in advance

Options history reaches back to 2022, so we use three non-overlapping windows. For H-B: **discovery 2022–23**, **confirmation 2024–25**, **final Jan–Aug 2026, run once**. A cell is "starred" when the 95% bootstrap CI of events minus placebo excludes 0. H-B passes discovery only if the CSP is positive and starred at ≥2 of the 10/21/42-session/expiry horizons. For H-A (registered earlier): in-sample 2024–25, OOS Jan–Aug 2026 once. Trial 9 (long straddle after exits, `HYPOTHESIS_9.md`) was tested once on 2022–23, a window never used for it. In total, 9 pre-registered trials are reported (`TRIALS.md`).

3. Results: cash-secured put, events minus placebo (% of spot), 3–6m options

Hypothesis · window	n	10	21	42	63	exp	Verdict
H-A exec exits · 2024–25	163	−0.32	−0.38	+0.10	−0.06	+0.11	≈ 0
H-A exec exits · 2026 (OOS)	58	+0.46	+0.66	**+1.62***	**+2.52***	+1.65	sign flip
stock only · 2026	55	+2.29	**+6.84***	**+8.97***	+8.43	+2.80	drives it
H-B CFO appt · 2022–23 (discovery)	34	+0.01	+0.00	−0.41	+0.65	+2.34	0 stars: fail
H-B follow-up, exp · 2024–25 (confirm)	30	−0.16	+0.25	+0.23	−0.41	−0.23	fail
Trial 9 long ATM straddle, 1m · 2022–23	54	−0.48	−1.15	–	–	–	fail (net of 4% cost)
CEO appt · 2022–23	14	+0.66	−0.31	+0.86	+1.36	+1.10	0 stars
CFO departure · 2022–23	25	+0.09	−0.32	−0.93	−1.34	−0.33	no edge, as predicted

* = 95% CI excludes 0. n = priced events at 21 sessions. Straddle 21d CI [−3.87, +1.42]; its 3–6m version: −1.00% (21d), −1.10% (42d). Exp CI for H-B: discovery [−0.31, +5.06], confirmation [−3.07, +2.14].

H-A fails. The put trade is about zero in 2024–25. In 2026 it turns positive only because exit stocks rallied +6.8–9.0% vs placebo, and a short put is long delta. That is regime luck, not the volatility

premium. **H-B fails.** It has no stars in discovery. The nearly-starred expiry cell (+2.34%) was registered as a follow-up (trial 8, `HYPOTHESIS_8.md`) and tested on unseen 2024–25: −0.23%. The near-star was noise.

Starred cells vs chance (taking the workshop's advice seriously). At our pre-declared key horizons, 2022–23 has **0 starred of 97** cells (3 tags × top-100/tech × 5 strategies × 4 horizons), against 4.9 expected by chance. Across the notebook's full boards there are 8 of 135, against 6.75 expected, all at 1–5 sessions. Stars appear at the chance rate, so reading individual stars as signals would be p-hacking. The confirmation step shows what happens to the best-looking one. Large-cap tech (tip): only 6 CFO-appointment events in 2022–23, positive but uninformative.

4. What replicated, and why it still cannot be traded

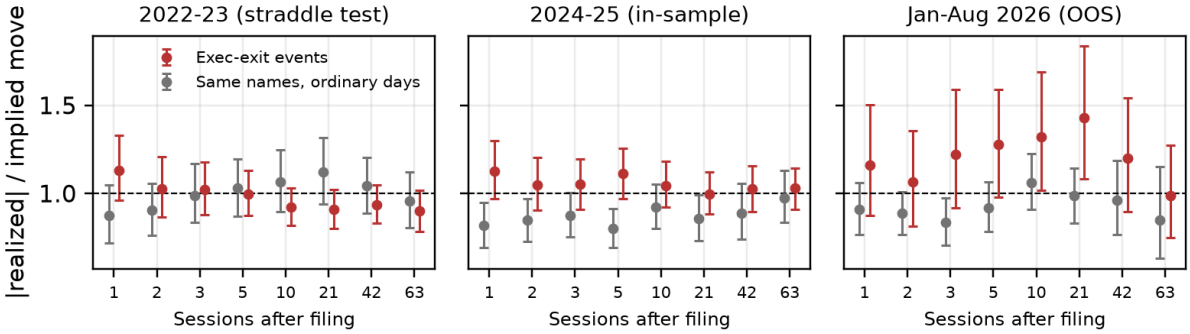


Figure. |Realized| ÷ implied move (3–6m ATM straddle on the pre-event session, √t-scaled), mean and 95% CI, in three windows.

In all three windows, the filing-day move is under-priced: the 1-session ratio is 1.13 vs placebo 0.88 (2022–23), 1.13 vs 0.82 (2024–25), and 1.16 vs 0.91 (2026). **The multi-week excess is not robust:** at 10–21 sessions it is 1.05–1.43 in 2024–26, but 0.91 vs 1.07–1.12 in 2022–23 (reversed). So the market under-prices the exit *surprise*, then reprices within a day. A buyer entering at the filing-session close (the first tradeable moment) is too late. Trial 9 confirms this: the post-filing straddle loses −1.15% vs placebo at 21 sessions, with a 31% hit rate. Selling puts does not pay either (0 of 54 starred CSP variants in the pre-declared sensitivity grid). Information in leadership 8-Ks is priced quickly: an honest, coherent null.

5. What would break it, how we would trade it, and our sealed-window predictions

Limits: small samples (14–163 priced events per window); a static 2026 top-100 list (survivorship); filing date ≠ announcement time (many exits are press-released first); daily last-trade marks, not NBBO mid. **What could still work (untested):** capturing the day-0 surprise needs entry *before* the market reprices. That means using EDGAR acceptance timestamps to trade the open after an overnight filing, or intraday data. Top-100 ATM options trade almost every session; at 1–2 contracts per event, ADV does not bind. **Hash-stamped predictions** (`FORWARD_LEDGER.md` P1–P4, `FORWARD_LEDGER_2.md` P5–P7): on your window, exit and CFO-appointment CSPs will not beat placebo, and stars will appear at about the chance rate. Our P2 (exit realized/implied > placebo at 21 sessions) already failed on 2022–23, and we report that against ourselves.